

A SMART WHISTLE COUNTER For Cooking And More

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This article describes a smart electronic counter that counts the number of pressure-cooker whistles and also time duration of the cooking. It triggers a buzzer alarm when the preset whistles count or preset cooking time duration is reached.

This design can also be used in factories having boilers with pressure release valves to automatically turn them off on hearing the sound at boiling point. Or it can be used in steam engines to automatically cut

off power when sound is made by the vessel's outlet valve.

Block diagram of the project is shown in Fig. 1. The components used in the project are listed under the Bill of Material table.

Circuit and working

Circuit diagram of the smart whistle counter is shown in Fig. 2. The circuit comprises Arduino Mega 2560 board (MOD1) and LM393 based sound sensor module (MOD2), which is used for detection of the whistle's sound.

It operates at 5V and produces an analogue voltage proportional to the sound level. This voltage is given to analogue input pin (A0) of microcontroller MOD1.

The 4x4 keypad is used as an input device to set the total number

of whistles required, or total time duration for cooking. The keypad has keys arranged in four rows and four columns, which are connected to eight digital input pins of the microcontroller.

The 23mm (0.91-inch) I2C serial pin 4 of the OLED display module is interfaced with Arduino. It displays the number of detected whistles and time duration. The resolution of the display is 128x32 and its driving voltage is 5V. It is connected to the SCL and SDA pins of the Arduino board.

Active alarm buzzer (B1) used in this project works on 5V. It is connected to digital output pin 53 of Arduino board MOD1. It provides 1kHz sound signal. The buzzer is triggered when the preset whistle count or the preset cooking duration is reached.

Software

The sound sensor MOD2's output pin 1, connected to analogue input A0 (pin 54) of microcontroller MOD1, is digitised with 10-bit resolution. A moving window low-pass filter is implemented in software with the window size of 25,000 samples. The sum value of window is compared with the 'high threshold' and a whistle is declared detected if the sum value is higher. The whistle sound stays high for a few seconds.

BILL OF MATERIAL		
Component	Part Number	Quantity
Microcontroller board	Arduino Mega 2560 (MOD1)	1
Sound sensor	LM393 (MOD2)	1
Display	23mm OLED 128x32 (MOD3)	1
Keypad	4x4 keypad (MOD 4)	1
Buzzer	MH-FMD (B1)	1
On off switch	S1	1
Battery	9V battery	1

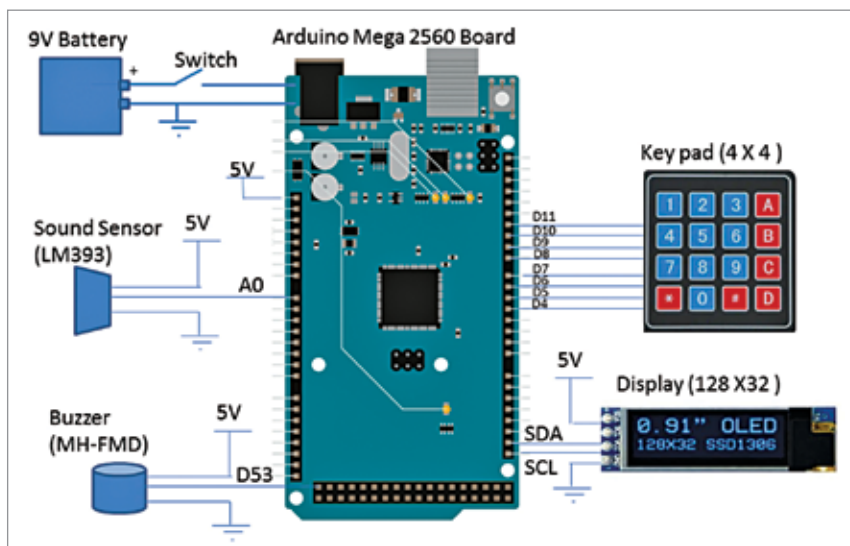


Fig. 1: Block diagram of the project

EFY Note

The source code of this project is available for free download at www.electronicsforu.com